



ÜSKÜDAR UNIVERSITY

Department of Computer Engineering

# *A Semantic Score Generation Method for Health Misinformation Detection*

**Presented by**

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## ! Problem Definition

### Days

Online health claims circulate at scale, but verification is slow, manual, and expert-bound.

### LLMs

Large Language Models alone are confident liars: fluent generation without grounded evidence produces high-credibility hallucinations on medical topics.

### 1 KB

Existing <sup>1</sup>Retrieval-Augmented Generation(RAG) systems use a single knowledge base — they verify what looks true but never see what looks false.

## 🎯 Motivation & Objective

Can a <sup>2</sup>dual-corpus GraphRAG pipeline outperform single-corpus baselines on health claims?

### G1: Ground

Anchor every prediction in verifiable evidence retrieved from two domain knowledge graphs.

### G2: Discriminate

Encode a signed distance to truth versus falsity as an explicit, learnable feature.

### G3: Balance

Recover minority-class performance under natural class imbalance without losing accuracy.

### G4: Explain

Surface the linguistic and graph signals driving each decision for human review.

## Methodological Innovations



### Dual-KB Retrieval

Evaluates claims against verified facts and fake news simultaneously. Subtracting these similarity scores yields a metric directly quantifying proximity to truth.



### Semantic Health Graph(SHKG) & PageRank

PageRank calculates the 'authority' of each term to generate a Semantic Score, verifying the structural reliability of medical concepts beyond mere text.



### Improved Hybrid Feature Vector

Integrates semantic, graph, and linguistic patterns (hedge, sensational, numeric) into a unified, interpretable foundation for classification.



### SVM-SMOTE & Balanced Analysis

Synthesizes minority samples at the decision boundary (Tomek Links) and measures how data skew impacts detection across balanced/imbalanced splits.

**PageRank:** Algorithm ranking the centrality of a concept.

**SHKG:** Maps structural relationships between medical entities.

**Tomek Links:** Data-cleaning method removing confusing cross-class points.

# Where Prior Systems Fall Short

## Large Language Models

GPT-3.5, LLaMA 3.2, PaLM 2, Claude 3.5, Gemini 1.5

- ✗ No retrieval grounding
- ✗ Confident hallucinations
- ✗ Closed-source, costly API
- ✓ Strong language understanding

## Single-KB GraphRAG

Standard Retrieval-Augmented Generation over one corpus

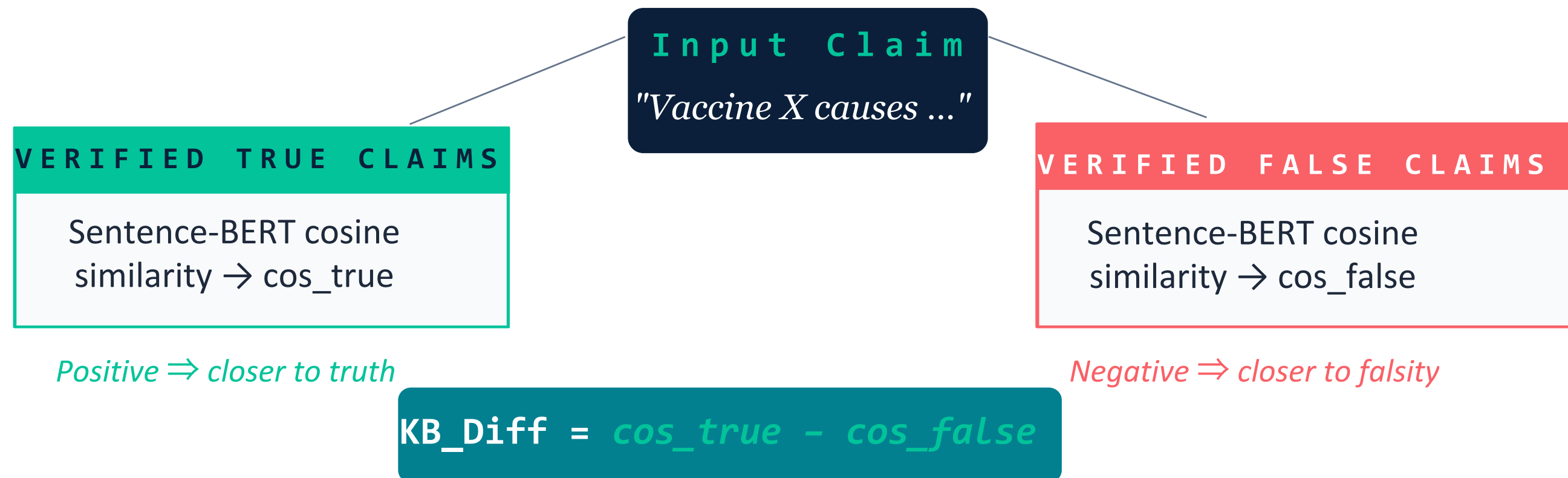
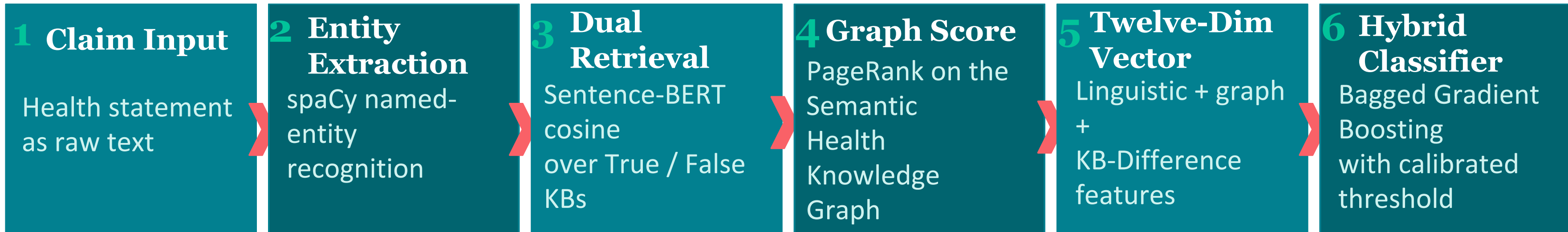
- ✓ Retrieval-grounded answers
- ✗ Cannot model what is false
- ✗ No graph-topology features
- ✗ Ignores class imbalance

## Dual KB (Ours)

Dual Knowledge-Base GraphRAG + SVM-SMOTE + Tomek Links

- ✓ Dual True / False retrieval
- ✓ PageRank graph features
- ✓ Open-source, reproducible
- ✓ Imbalance ablation studied

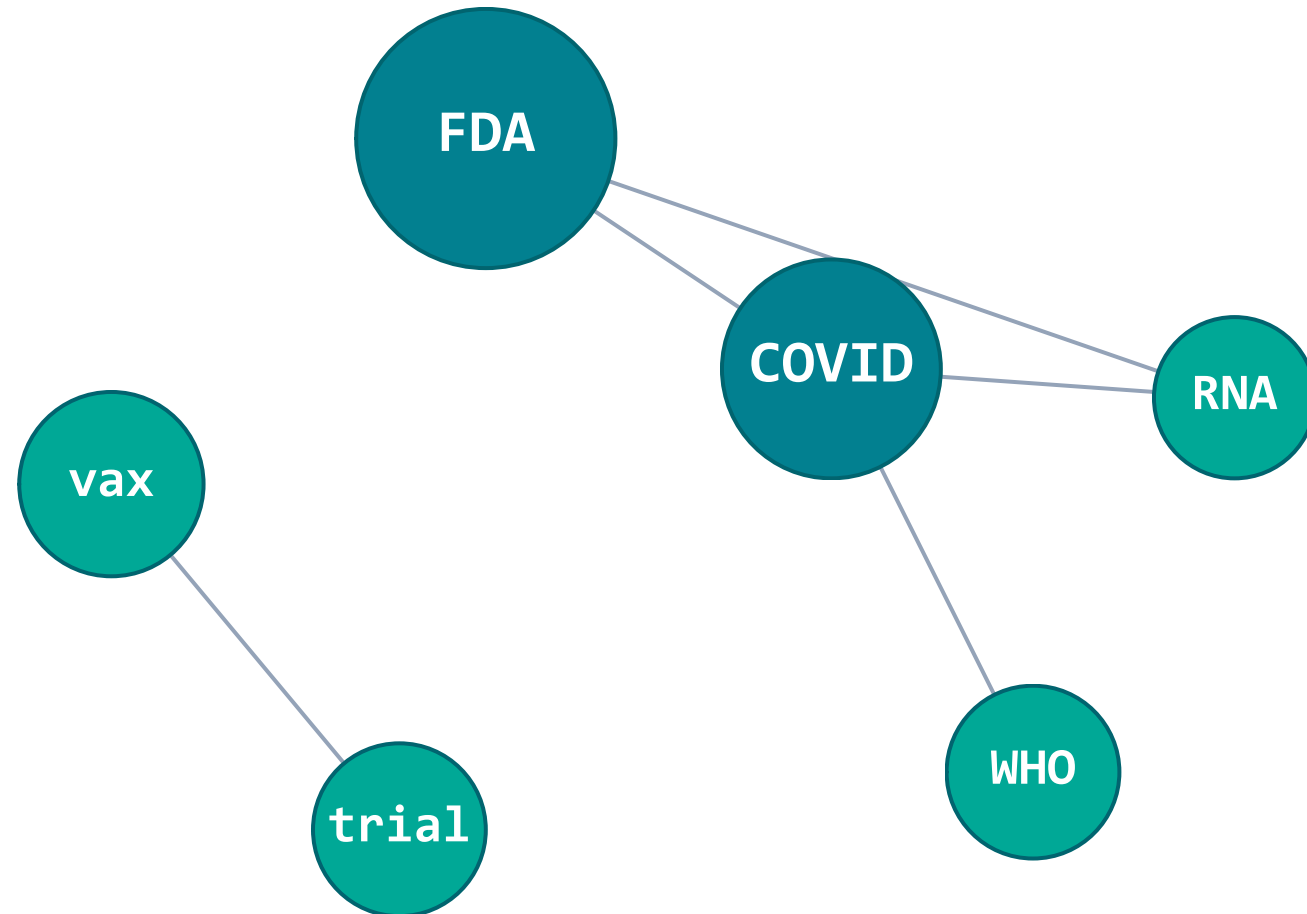
# End-to-End Pipeline



**Sentence-BERT cosine:**  
quantifying the vector distance between input claims and knowledge base embeddings.



## Semantic Health Knowledge Graph (SHKG)



**Nodes:** medical entities extracted from training claims via spaCy named-entity recognition.

**Edges:** co-occurrence of two entities in the same claim, weighted by frequency.

**Centrality:** PageRank with damping factor  $d \in \{0.6, 0.7, 0.85, 0.95\}$ .

**graph\_score:** mean PageRank of the claim's entities — injected as feature 7 in the 12-D hybrid vector.

## Resampling

How we fix the natural health claim imbalance:

### STEP 1

#### Before

*Natural 2 : 1 imbalance*

**2 : 1**

- True majority class dominates the decision space.
- False minority class clusters loosely around overlap region.

### STEP 2

#### + SVM-SMOTE

*Boundary-aware synthesis*

**1 : 1**

- A Support-Vector-Machine identifies the minority decision frontier.
- Synthetic False samples are generated only along that frontier.

### STEP 3

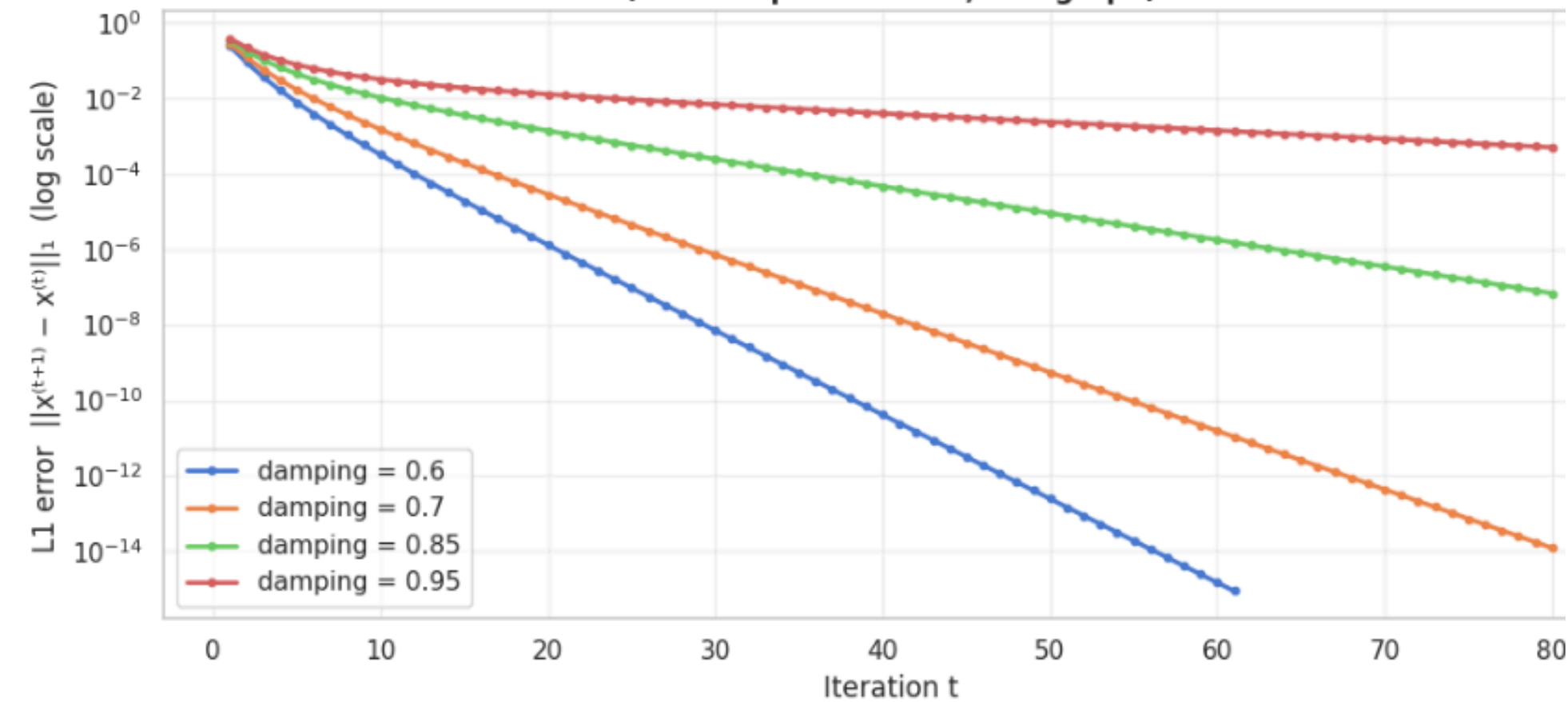
#### + Tomek Links

*Overlap removal*

**✓ Clean**

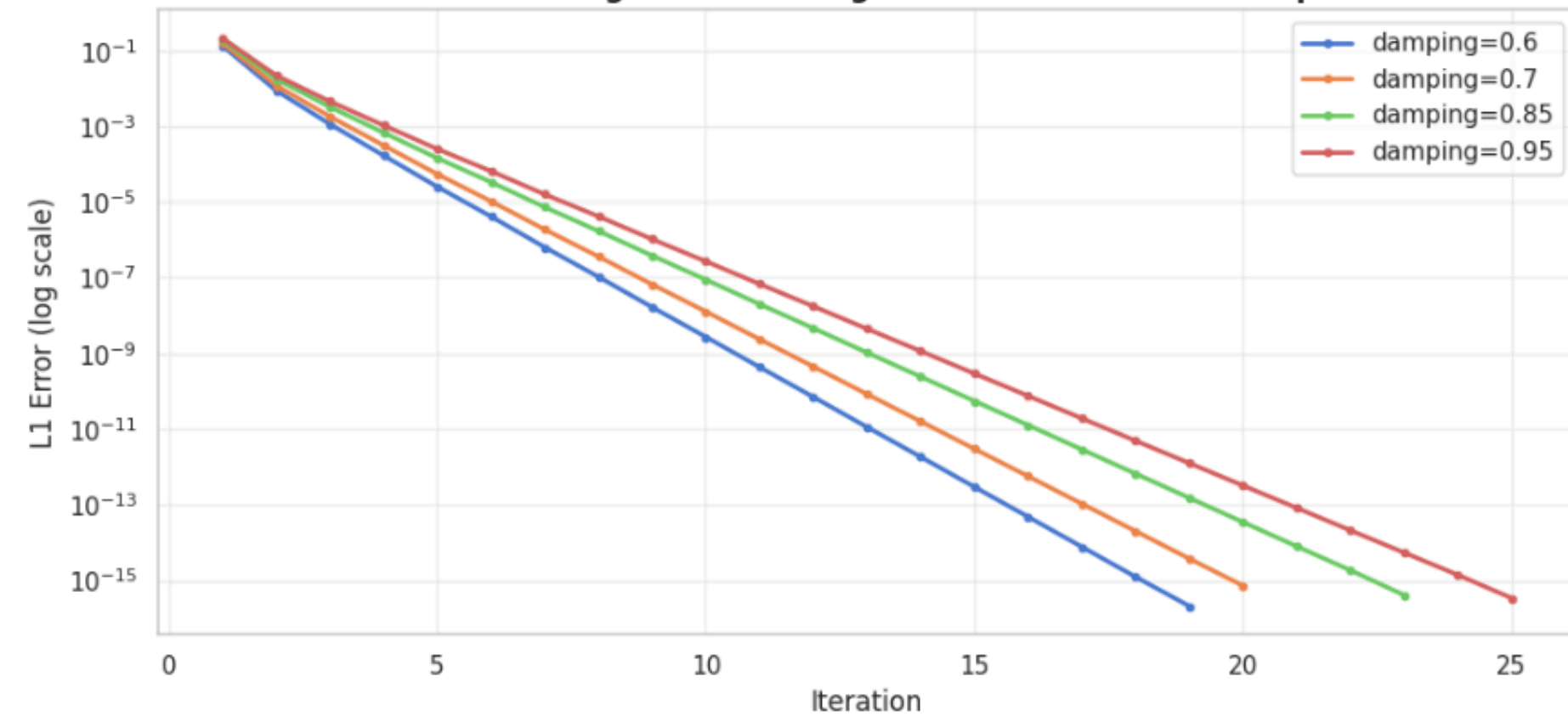
- Cross-class nearest-neighbour pairs are detected.
- The majority-side member is removed to sharpen the boundary.

Measured PageRank Convergence on SHKG  
(L1 error per iteration, real graph)



TST graph nodes: 40  
TST graph edges: 679

Measured PageRank Convergence on TST Semantic Graph



### SHKG: Contextual Density

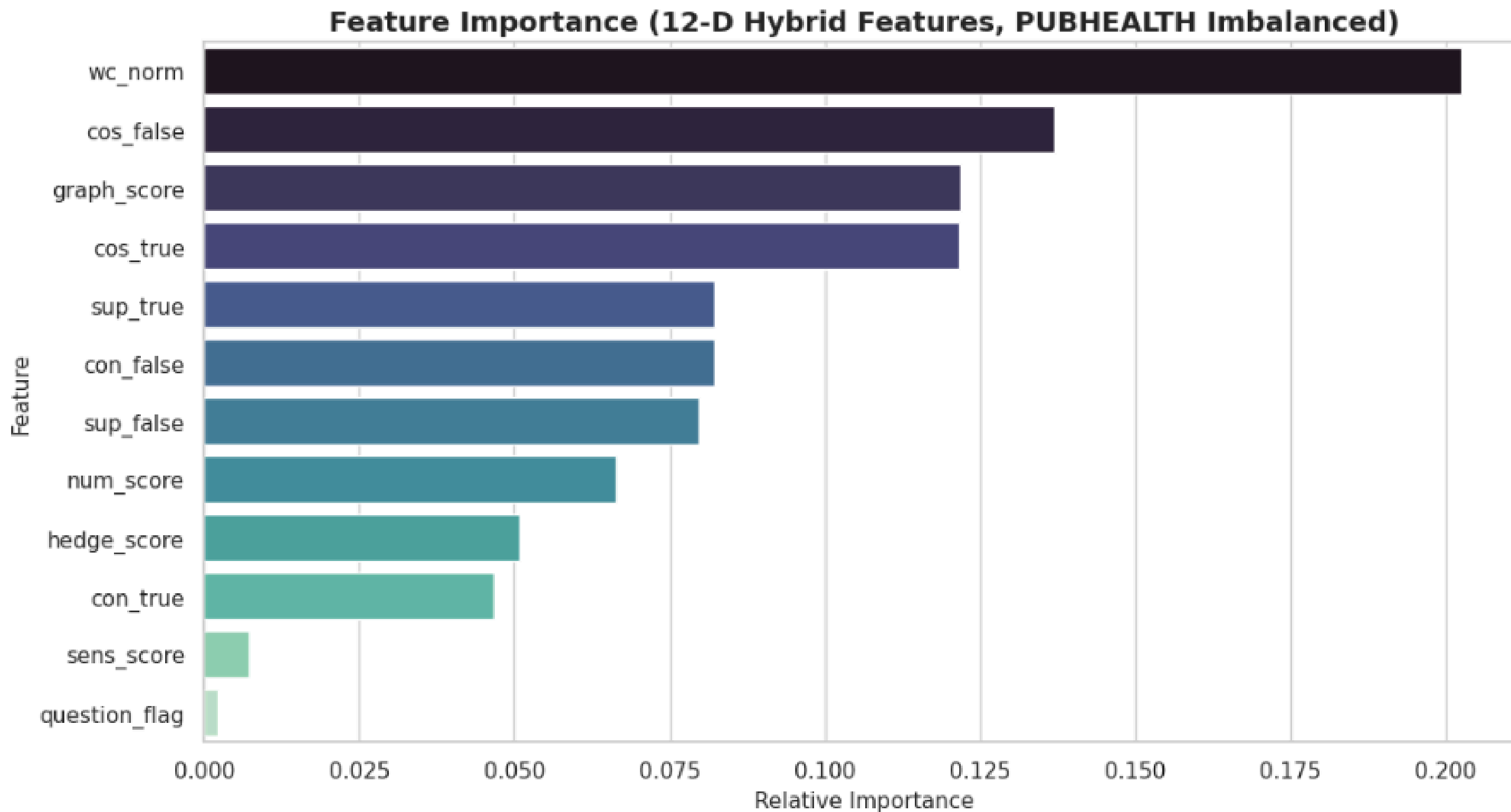
- **Convergence:** Gradual (60-80 iterations).
- **Topology:** Dense; preserves indirect medical relationships.
- **Strength:** Superior semantic depth without information loss.

While the reference TST model maximizes speed through structural sparsity, our SHKG model prioritizes contextual data integrity.

### TST: Structural Sparsity

- **Convergence:** Fast (~25 iterations).
- **Topology:** Pruned and optimized for low latency.
- **Strength:** Maximum computational efficiency.

## Entity Coverage and KB-Proximity Distribution

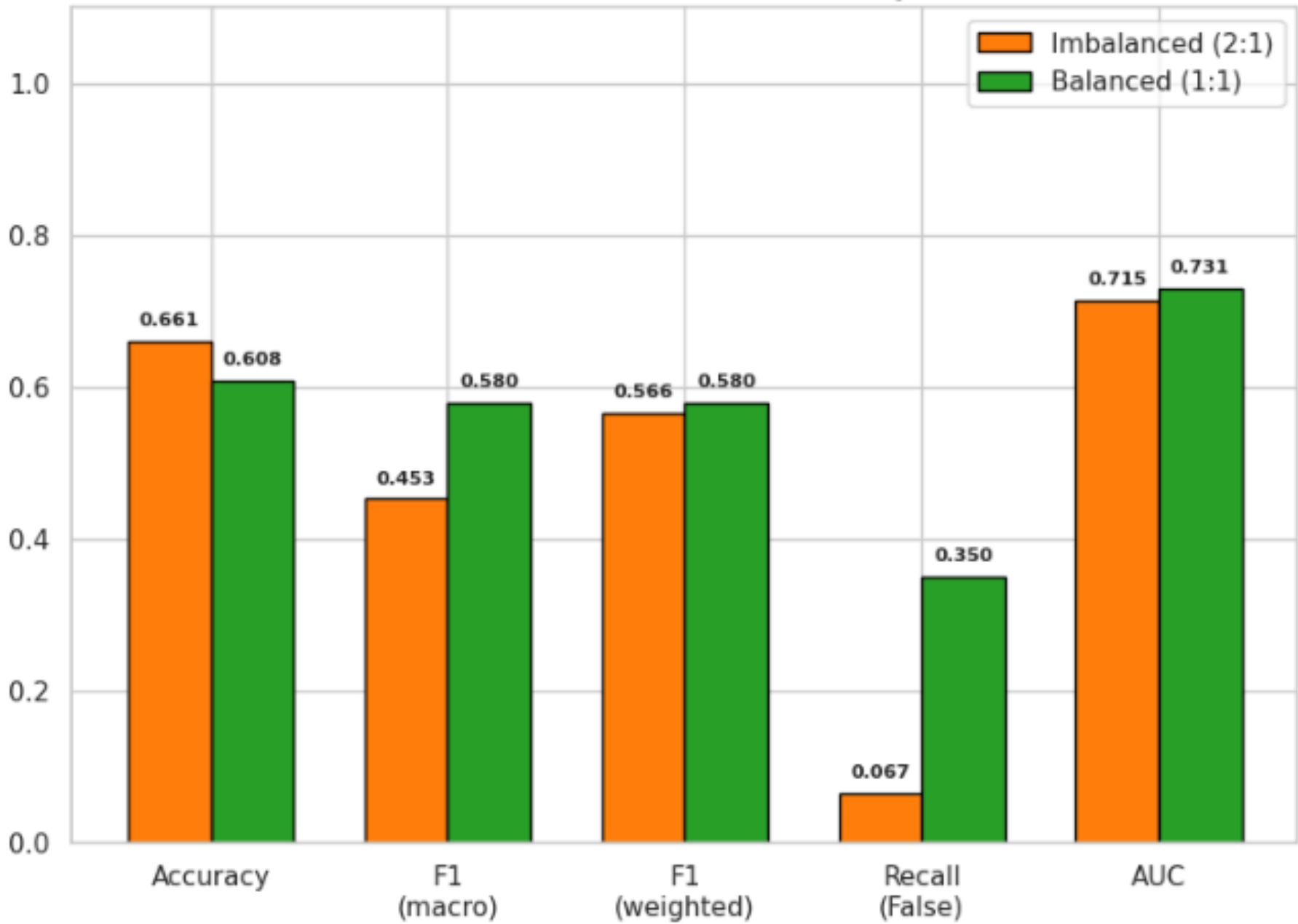


Semantic similarity and graph-based features provide the strongest discriminative signals in our architecture. These features deliver highly discriminative signals beyond the capabilities of prior Single-KB systems.

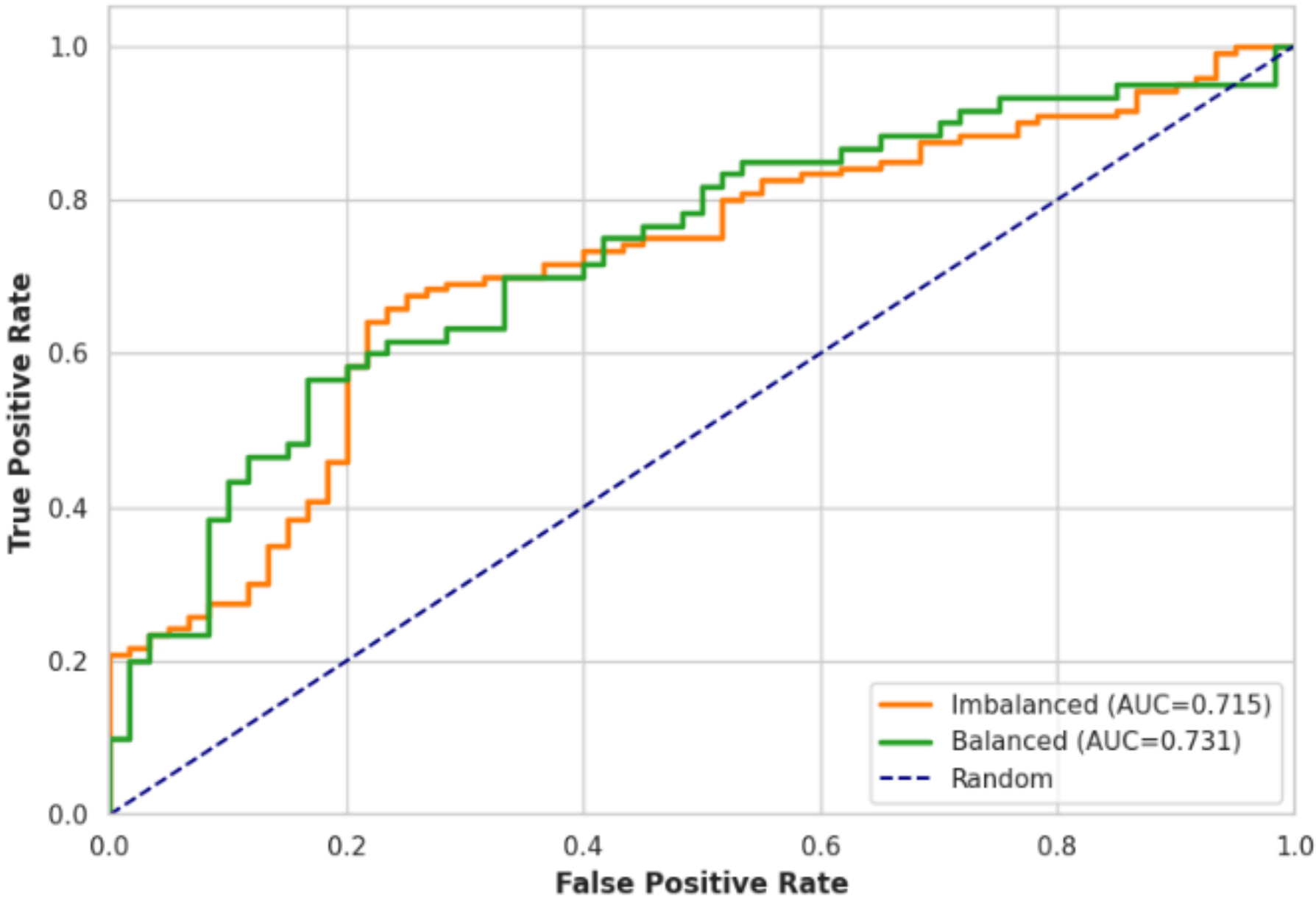


| Setting                    | Train size | Test size | Accuracy | F1 (macro) | F1 (weighted) | Recall (False) | Recall (True) | AUC-ROC |
|----------------------------|------------|-----------|----------|------------|---------------|----------------|---------------|---------|
| PUBHEALTH Imbalanced (2:1) | 720        | 180       | 0.661    | 0.453      | 0.566         | 0.067          | 0.958         | 0.715   |
| PUBHEALTH Balanced (1:1)   | 480        | 120       | 0.608    | 0.580      | 0.580         | 0.350          | 0.867         | 0.731   |

Imbalance Ablation — Metric Comparison



ROC — Imbalanced vs Balanced



- Balanced sampling reduces majority-class dominance and improves minority-class detection. Macro-F1 increases from **0.453** → **0.580**, while False-class recall improves **5.2× (0.067→0.350)**.
- A small drop in accuracy is traded for a more balanced and reliable classifier. The improved AUC (**0.715** → **0.731**) shows that SVM-SMOTE enhances overall discrimination performance.



# Benchmarking on PUBHEALTH

| Method                                           | Dataset               | Resampling             | Accuracy | Precision | Recall | F1 (weighted) | F1 (macro) | Recall (False) | AUC-ROC | Notes                              |
|--------------------------------------------------|-----------------------|------------------------|----------|-----------|--------|---------------|------------|----------------|---------|------------------------------------|
| TrumorGPT (TST+Centrality+Jaccard)               | Balanced (1:1)        | No                     | 0.500    | 0.250     | 0.500  | 0.333         | 0.333      | 0.000          | 0.491   |                                    |
| TrumorGPT (TST+Centrality+Jaccard+SMOTE+Bagging) | Imbalanced (2:1)      | SVM-SMOTE + TomekLinks | 0.328    | 0.165     | 0.492  | 0.165         | 0.247      | 0.983          | 0.496   |                                    |
| Novelty Method (Dual-KB)                         | Balanced (1:1)        | No                     | 0.608    | 0.648     | 0.608  | 0.580         | 0.580      | 0.350          | 0.731   |                                    |
| Novelty Method (Dual-KB+SMOTE+Bagging)           | Imbalanced (2:1)      | SVM-SMOTE + TomekLinks | 0.661    | 0.558     | 0.513  | 0.566         | 0.453      | 0.067          | 0.715   |                                    |
| TrumorGPT GPT-4 (paper original)                 | PolitiFact 600 claims | -                      | 0.885    | 0.914     | 0.850  | 0.881         | -          | -              | -       | different dataset -- referans only |

TrumorGPT vs Novelty Method | PUBHEALTH

*\*Paper original uses a different dataset and costly closed-source API, presented for reference only.*

- The proposed Dual-KB novelty architecture improves minority-class detection while maintaining competitive overall performance.
- Class balancing reduces majority-class dominance and increases False-class recall from 0.067 to 0.350.

📄 Baseline results sourced from:  
**Hang, C. N., Yu, P.-D., & Tan, C. W. (2025). “TrumorGPT: Graph-Based Retrieval-Augmented Large Language Model for Fact-Checking.” *IEEE Transactions on Artificial Intelligence*, 6(11), 3148–3162.**

*A misinformation classifier is sustainable only if it is lightweight to train, reproducible by anyone, and auditable for end-users — not just accurate on a benchmark.*

## L I G H T W E I G H T

### Single-Colab Footprint **12-D**

*feature vector*

A 12-dimensional feature vector trained on 720 claims runs end-to-end on a free Colab session — no multi-GPU cluster, no special hardware.

## R E P R O D U C I B L E

### Open-Source Stack

**0 \$**

*inference cost*

Sentence-BERT, spaCy, NetworkX, scikit-learn, imbalanced-learn — every dependency is open and pinned. Closed APIs drift in cost and output; open models do not.

## A U D I T A B L E

### Per-Prediction Explainability **12 / 12**

*features inspectable*

Feature importance is computed across the bagged ensemble; reviewers can inspect which signals — KB\_Diff, graph\_score, hedge\_score — drove each verdict.

# Thank you

Questions are welcome